

Digital Farming: The New Era of Agriculture-Opportunities and Challenges

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Abstract

Agriculture has its direct effects on the economy of the country, farmers' income, and the population who lives there. Therefore, an appropriate approach to do effective farming is required so that farmers have optimum results from their crops and cultivation. Digital farming enables the farmer to do so by using different tools and techniques. By using drones and sensors, farmers can spray pesticides, grid sampling can be applied to check the fertility of soil like using the global positioning system, yield monitors can be used to monitor the crop yield from time to time and feed the same data in software for later analysis and comparison, variable rate technology is also one of

the techniques to control the rate of fertilizers, etc., and different precision farming techniques can be applied. The current pandemic situations raised the requirements of digital farming and the need for relative education on digital farming. In this chapter, the opportunities and challenges of digital farming are discussed: how digital farming techniques will help the nation or farmers grow their economy in a sustainable manner and what are the real problems that become challenges for farmers to accept the concept of digital farming.

Keywords: Digital farming, techniques and tools, precision farming, opportunities, challenges, smart farming monitoring system services (SFMSS)

3.1 Introduction

The conventional way of doing agriculture can be characterized as a basic style of yielding. It holds the intensive use of traditional tools, manpower, and prior information on farming. The use of traditional tools, natural resources, organic fertilizers, and cultural beliefs is very usual when we talk about old ways of farming. Here farmers do farming based on the principles of mixed farming, with traditional ways and without using any artificial techniques. Traditional farming usually involves small plantations established according to local

tradition. It mainly uses organic manure. Traditional farming uses organic manures, synthetic chemicals, and fertilizers, which are usually genetically modified, to get the best out of a particular crop or crops. Here significant amounts of organic and dynamism inputs are required in this method that undermine a kind of ecology. It is still used by about 50% of the world's population. In traditional agriculture, irrigation is also done in a traditional way, in which some traditional methods of irrigation include the basin method, drain irrigation method, and strip irrigation method.

Along with having some advantages over traditional farming methods, it also has some drawbacks compared to digital farming. Some of these advantages/drawbacks are as follows:

- Crops are purer without manure: These crops are fresh and are healthier and more beneficial.
- Crops without manure can be sold at a higher price.
- Crop waste after decomposition can be used again as a fertilizer for soil.
- In traditional farming, farmers must spend about 15 h to harvest the crop, which is much more than in high-tech farming.
- Soil is used in traditional farming due to which decomposition takes longer time. In this, the crops are very

prone to soil diseases.

- Crops take longer time to harvest, so crops are becoming expensive to earn the profit.
- Pesticides are used to protect crops from pests, which make the plants not so healthy.
- Plants are not well nourished in traditional farming because the roots do not penetrate deep enough into the soil to absorb water and minerals.
- Traditional farming is heavily dependent on rainfall to water the plants. In this, the quantity of crops produced per acre is reduced, which causes starvation in the event of drought.

To overcome the drawbacks of old farming maneuvers, new technologies can be welcomed. Internet of Things (IoT) and artificial intelligence play the main role here to automate most of the farming decisions. IoT-embedded tools can be used for soil monitoring and water management [\[12\]](#). The combination of IoT with cloud computing makes the farmer high-tech to reduce human-made errors by replacing human-based decisions with computer-based decisions to make economic benefits [\[11\]](#).

3.1.1 Digital Farming

Digital farming transmits quality agriculture information to appropriate locations, methods, and decisions to illuminate and predict farming issues across all farms. An important part of digital farming is timely access to quality information and decision making. Digital farming through digital technologies has the potential to provide the farmers with the knowledge and capabilities to meet the challenges related to farming and seize growth opportunities. Through digital farming, agricultural products can be made better, and its efficiency can be enabled. It provides an increase in value for the farmers. Digital farming is a facsimile of the application of modern information and communication technologies in the field of agriculture. Digital farming aims to form the basis of a decision-making support system for agricultural management. Digital farming provides useful information about crop growth, climate change's use in farming, and labor issues. It makes crops useful by attracting a lot of technical attention from engraining and watering to healthiness and collecting.

3.1.2 Automated Farming

Alterations in the global aging demographic have led to the adoption of automation in agricultural practices. Automation

and control system manufacturers have felt this change in the agriculture industry. With this change, the manufacturers have seen their sales increase as well. In the last few years, agricultural robots have also been included in agricultural work. Agricultural robots select the soil and crops according to their needs and reduce the need for manual labor. UAV/drone also generated the highest revenue among all the modern technologies used in smart farming. Most of the robots are used for deployment crop management. Planting seeds has become easier than a laborious process with automatic seed sowing machines in agriculture. With these automatic sowing machines, sowing can be done on the land much faster, which also saves time. Planting the seed at the correct depth and spacing the plants at the proper distance for optimum growth are two factors required for effective seeding [3]. [Figure 3.1\(a and b\)](#) shows the process of sowing seeds.

Apart from sowing seeds, an automated process has a watering and irrigation method that permits agriculturalists to rein when and how much water their harvests get. Weeding and pest control are also both delicate aspects of plant being looked after, and these farm duties are perfect for self-sufficient automata. The basic perception of integrating self-directed farming into farming is to increase efficiency, product yield, and quality, because of which the dependence on manual labor can be

reduced. We can say that by using autonomous farming, the farmers of the future will expend their phase doing tasks like overhauling apparatus, debugging robotic coding, investigating data, and scheduling farm operations. In contrast, the ancestors of the peasants spent most of their time in heavy labor.



(a)



(b)

Figure 3.1 (a) Manual process of sowing seeds. (b) Automated process of sowing seeds [3].

3.2 Tools and Methods Used in Digital Farming

There are several tools that can be used in digital farming to help farmers not only in the cultivation process but also in planting and many more. In the past 10 years the use of these tools has increased. The tools that are used in digital farming are discussed below.

3.2.1 GPS

The Global Positioning System (GPS) is known as a precision farming technique, where farmers can remotely control their farming in case of bad weather, darkness, downpour and fog, etc. GPS locates the position of a farmer's lands or the machines in a farmer's lands with the help of satellites. Satellites can track the farmer's lands with the help of meters that are installed on land. GPS helps to view any part of land from a remote location, and thus farmers can maintain their lands, crops, and other related things. GPS techniques also make the farmer sprinkle pesticide and fertilizers by mapping the location with their home screen. In conventional farming, Global Positioning System-based solicitations are being used mostly for farm planning, field mapping, tractor guidance,

variable rate applications, soil sampling, and yield mapping. Through Global Positioning System, farmers get to know about the small reflectivity area surroundings like rain, dust, fog, and darkness, which helps them during their work.

3.2.2 GIS

The Geographical Information System (GIS) works in combination with GPS. They both complement each other. It is a software application that is used to store and maintain information or feedback data from GPS. Thus, GPS can be used to collect information, and that information can be analyzed with the help of GIS.

3.2.3 Grid Sampling

Grid sampling is a technique where the land is divided into small chunks and the soil from these chunks are sent to the lab for analysis. Therefore, we can find which type of soil from a particular grid is good for which type of crop.

3.2.4 Variable Rate Technology

Variable rate technology is a technique that is used to control the amount of seeds, pesticides, fertilizers, and other things that

are applied in the field. Thus, with variable rates, we can control the application of these things in our fields.

3.2.5 Yield Monitors

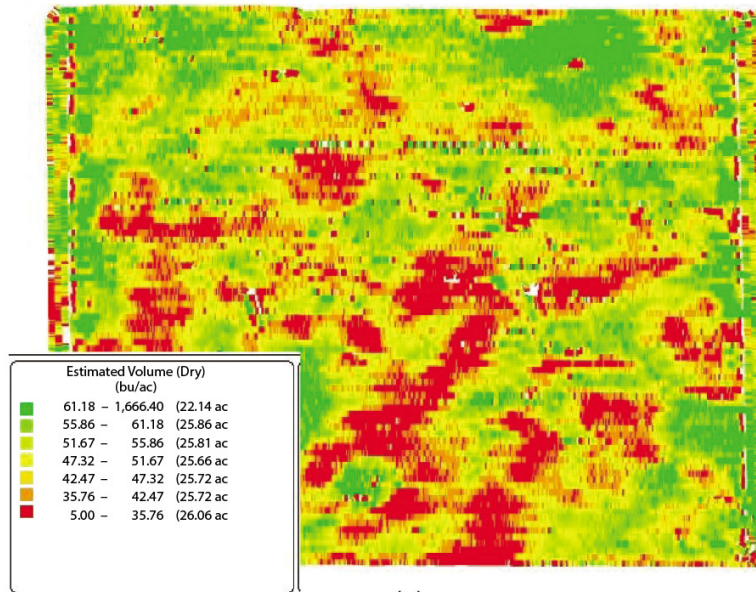
Yield monitors are the devices which are used to collect data about crops. These devices are used by farmers to improve the efficiency of their crops and to reduce cost [2]. By using these devices, farmers can track the moisture of grains and collect data about crop mass flow tons per hectare. Yield monitors are used to record the crop yield and moisture. After collecting the data, these records are displayed in the form of a map in the monitor as shown in [Figure 3.2\(a\)](#).

3.2.6 Remote Sensors

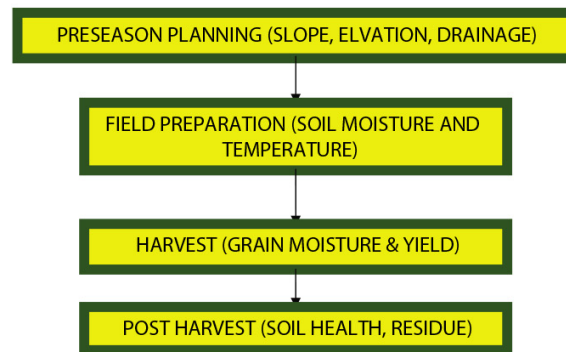
Here sensors are used to track the crop as well as the field being far away from the field. The sensors that are used can be aerial or satellite sensors. Sensors sense the changes on the basis of different colors that appear in the field such as soil color, crop development color, water, field boundaries, and so on. On the basis of colors, these analyze the status of crop- and field-related terms, and the farmer can see these records on the software or monitor screens which will be connected with that sensor. The information that is transferred by the sensor is done

by electromagnetic radiation (EMR). Some of the applications of using a remote sensor (RS) are given in [Figure 3.2\(b\)](#):

1. RS are used in decision making at the time of preseason planning. It includes decisions related to crop type, field operation schedule, amount of seeds for planting, etc.

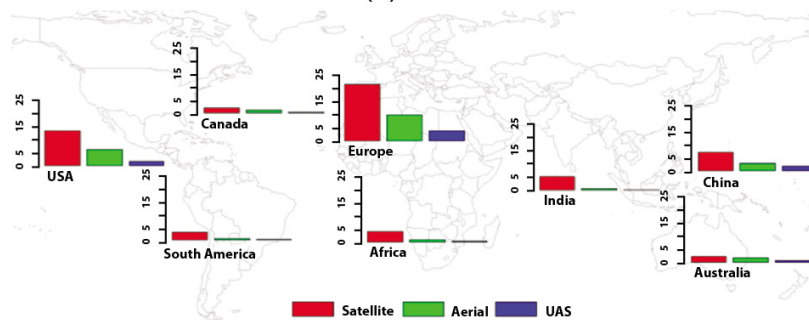


(a)

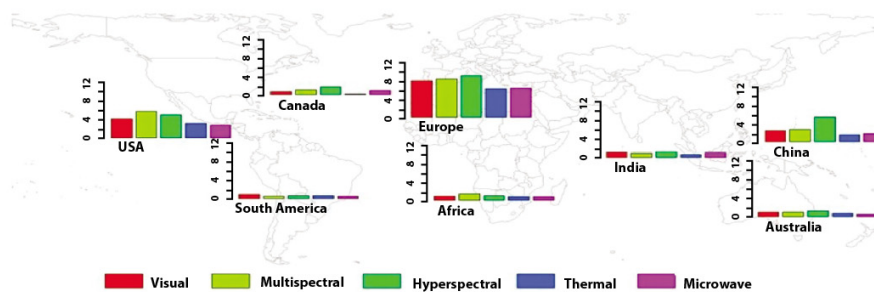


(b)

(a)



(b)



(c)

Figure 3.2 (a) Soybean yield field [2]. (b) Stages of production agriculture [10]. (c) Charts of spatial resolution of satellite sensors used since 1999 [10].

2. RS are used in field preparation, as crop germination takes place when the soil moisture content is according to the crop. Thus, it will help to analyze the moisture level of soil so that fertilizers can be used at the right time for better results. The land surface vegetation index [4, 10, 13, 14] is the term used to make decisions about field preparation tasks. Optical and thermal remote sensing data is used to find out that index.
3. RS are used in the stage of planting as well. Crop planting means maintaining the track of crop density that defines the emergence of crop and population of crop for knowing the harvesting logistics. If this data is available at the correct time, then the farmer can use it to do replanting at an accurate time without losing the correct time span or, we can say, without losing the correct season or time to catch the growth. It also helps to take other crop-related decisions such as herbicide, fertilizers, and pesticides used in mid-season so it will result in a good and healthy harvest.
4. RS is used in seasonal applications to find out crop stress due to pest and weed infestation. By using RS, one can see if there is any attack of pathogens on their crop or not, and if yes, then they can take precautionary measures to save their crop. For eliminating weed, herbicide sprinkling is used but

before RS technique the spray of herbicides is done all over the field, which results in wastage of herbicides and increase in cost; however, with the use of high-resolution imagery, RS can be used to find out the patches where the weed is spread and only that portion will be subjected to spraying of herbicides.

5. Harvest means with the help of RS imagery the yield can be monitored. With the help of a yield map, yield density and variability can be estimated. This facility of early prediction of yield helps decrease yield-related losses.

[Figure 3.2\(c\)](#) shows the different types of satellite used from 1999 by different countries, and their use rapidly increased and spread over the globe. [Table 3.1](#) shows the readings collected from remote sensors about yield assessment. RS is used to assess spatial and temporal data.

6. RS is also used in the post-harvest process and can be used to make decisions on residue. Crop residue management is important to continue the effectiveness of the soil of that field.

Table 3.1 Remote sensor data studies for crop yield analysis [10].

Spatial/temporal resolution	Platform/sensor/data	Accuracy compared to <i>in situ</i> data
0.02 m; three-time acquisition during early and midseason crop development	Visual sensor on board UAS	R ² of up to 0.74
30 m; imagery for 2008 to 2013	Multispectral sensor; Landsat 5 and 7	R ² = 0.14 to 0.58 for corn and 0.03 to 0.55 for soybean yield estimation
0.3 m; bare soil imagery	Multispectral sensor onboard a manned	R ² = 0.52 to 0.97 for

Spatial/temporal resolution	Platform/sensor/data	Accuracy compared to <i>in situ</i> data
	aircraft	corn yield estimation

3.2.7 Auto-Guidance System

This system is used to perform symmetry while performing field operation as seed roping in a straight row and returning to the same row.

3.3 Pandemic Effects on Traditional Farming

The traditional method of agriculture involves labor and manpower from start to end. Thus, in the COVID-19 pandemic, the process of agriculture was affected badly. Around 1 billion people from all over the world get their livelihood from agriculture. They are the ones who share a large chunk of the economy also. Lockdown was the measure or preventive method used by the government to control the spread of COVID.

However, due to lockdown, the farmers become helpless to go to their field and continue the process required according to the cycle—for example, many farmers did not get access to go to the market and buy new seeds or fertilizers for their crop. The agriculture process has many steps which complete the whole process of farming, such as seeding, planting, nurturing, harvesting, and shipment to markets [5] that almost involve human labor in every step but which gets hindered due to COVID.

Low supplies and attacks of weeds and pathogens destroyed the crop, and due to impositions on transportation, herbicides and pesticides cannot be purchased and sprayed on yield at the time, thus resulting in reduced yields.

Due to traveling restrictions imposed in the nation, labor becomes unreachable to field and the production cycle gets disturbed. Reverse migration in COVID also will be a reason for disturbed agriculture at that time.

Isolation, lockdown, and self-quarantine are the key factors used during the COVID-19 spread that become useful, but due to so much restriction, the required procedure in agriculture become vanished and result into loss of yield or reduced production.

Due to COVID, the production of that year decreased, and that hiked the prices of items due to low availability and high demand as represented in [Figure 3.3](#). The prices of many vegetables (VP) that hold vitamin C were noticed to have the highest raise due to the pandemic.

The pandemic effect on traditional farming is that the production cycle was affected by the pandemic badly, and the prices of vegetables or other agricultural products become high as shown in [Table 3.2](#). The livelihood of many farmers gets effected, and so the economy of the nation was also deprived. Therefore, automated agriculture is needed, and its importance came to front so that possible steps of agriculture can be performed by using tools without the need of manpower.

Price	Weighted vegetables			Pak Choi		
	2020	2019	Decline (%)	2020	2019	Decline (%)
Feb	4.39	4.59	4.36	2.35	3.38	30.47
March	4.22	4.82	12.45	2.31	3.44	32.85

3.4 New Scope at Digital Farming

Every agricultural enterprise has complex equations of inputs and outputs. As some of the agricultural inputs are directly handled by the farmer such as selection of a crop, there are some environmental aspects also that are out of the farmer's control, such as meteorological conditions, typical weather, soil, and water accessibility which are sometimes beyond the control of the farmer. The unexplained effects of digital disruption hold the promise of potential profit. Digital farming refers to a connected, i.e., knowledge-based, agricultural fabrication scheme that employs intellectual networks and data-controlling utensils. It permits the mechanization of ecological procedures in farming.

In the section above, we saw how pandemic affected the production by disturbing agriculture procedure and also affecting the production cycle, farmer's lives, and economy. Thus, searching of all possible methods that can apply to reduce the loss is used and analyzed. In this new use of technologies, we get focused to use, i.e., IoT, cloud computing, sensors, drones, and artificial robots or machines.

From [Figure 3.4](#), we can see how the technologies can be used with farming to make it more automatic and machine-driven.

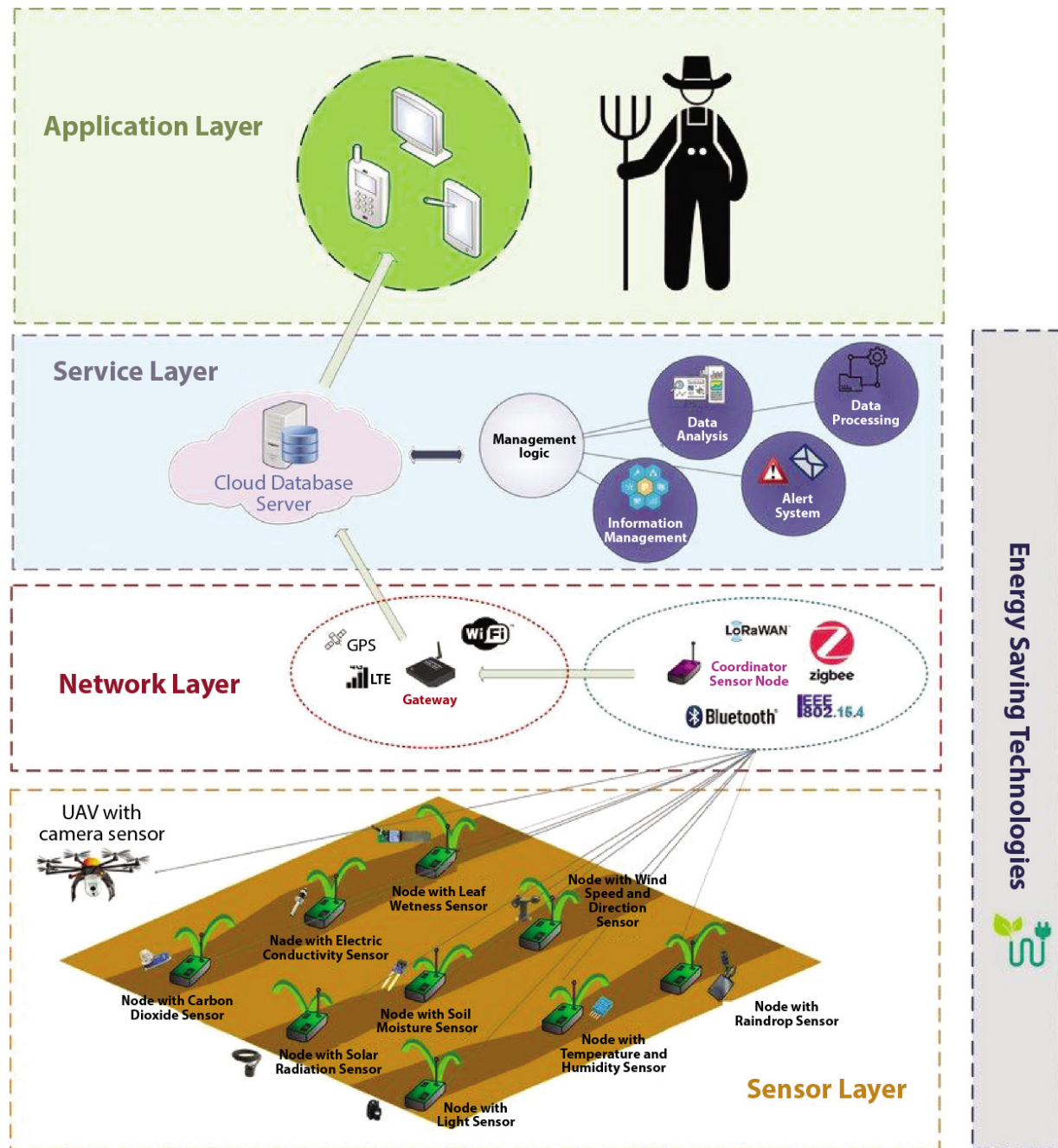


Figure 3.4 Architecture of precision farming [1].

The first layer is known as the sensor layer; the main work in precision farming is to sense the field on different factors. Sensors that can be used for sensing the pH value of soil are

known as electrochemical sensors; location sensors are used to sense the exact location of field longitude, latitude, and altitude [1]; and weather sensors are used to sense factors related to weather which influence the yield. There are factors mainly sensed by these sensors.

There is a different sensor that is used in agriculture for collection as given in [Table 3.3](#) and the analysis of different data to be required. These are mentioned below:

- Moisture and temperature of land
- Air humidity and flow

Table 3.3 Precision crop growing sensor models [1].

Sensor type	Sensor model
Soil moisture sensor	10-HS, SY-HS-220, FC-28
Temperature sensor	LM35, SHT15, DS18B20
Humidity sensor	DHT22, DHT11
Electric conductivity sensor	DFR0300
Wind speed and direction sensor	SEN0170
Barometric pressure sensor	BMP180
Carbon dioxide sensor	CDM4161A, MHZ16
pH sensor	MCP1525
Light sensor	TSL2561, BH1750

Sensor type	Sensor model
Solar radiation sensor	6450 TSR

Thermal sensor

Thermo MAP

- Air flow and land wetness
- Rainfall
- Soil pH value

The second layer is network. In N/w layer, a communication channel is set so that sensed data can be transferred to a destination for further processing. So, we can say that IoT and cloud computing techniques worked here to transfer or save the data.

The next layer is service layer, where the actual work on data begins. The service layer is aimed to analyze the data which is sensed in the first layer and transferred by the second layer. Based on analysis, a further decision on crop plantation and management is done.

The last layer is the application layer where the visualization of decision can be done. What is the result of a particular decision

that can be viewed here? Thus, precision farming is the new scope of digital farming.

Digital farming is identical and the same with smart agriculture and digital agriculture. Therefore, we can say that digital agriculture expertise is a skill-based technologically advanced ideal which is a combination of digital trade and manufacture procedures as the tools mentioned in [Table 3.4](#). This model pointedly challenges agriculturalists to traditional farming practices that may require upskilling and constant change. Here it is very important for farmers to adopt the mindset of digital literacy. Recognizing learning and capacity erection in digital savvy is essential to fill the capacity gap among farmers.

Table 3.4 Smart farming monitoring system services [1].

Service type	Tools	Description
Information management	MySQL Database, management logic	It defines the procedure of assembling, loading, handling, and sustaining evidence in all its forms.
Big data analytics, i.e., BDA platforms	Mahoot, IoT	It defines removing, onslaught, renovating, displaying, and conception of data with an objective to reveal significant and valuable figures that can support in originating verdict and take decisions.
Data processing	Classification algorithms	Classification of facts so as to decrease the

Service type	Tools	Description
		extent of redundant statistics
Data mining	Hadoop, Apache Spark Framework	Orderly and chronological manner of recognizing unseen outlines and evidence in a huge dataset

3.5 Challenges or Difficulties for Digital Farming

In traditional farming, it was problematic for agriculturalists to link production practices and harvest incomes with land inconsistency. In digital farming, it is possible to use more appropriate pesticides, herbicides, and fertilizers and better control the spread of those chemicals. The key advantages in digital farming are reduced costs, producing higher yields, and building an eco-friendlier farm.

In traditional farming, the farm with the farmer is very small, and it generates very few data points. Digital farming requires millions of data points to be appropriately aggregated and

analyzed; computing, storage, and processing power come at a charge. Thus, there is still some distance to be covered here for fields in traditional farming. The exact financial assessments of the total of expertise per unit land/per capita farmer and the associated reserves/yield on investment are not yet obtainable in certain contexts. This makes it particularly hard for everyone to assess technology from the point of view of economic aspects. A challenge for digital farming is also to reduce the cost of technology so that it is available and affordable to small farmers. Thus, it is natural that the farmer's ability to invest financially in technology is very limited. Keeping this in mind, it is necessary that the technology which is affordable to the farmers will be attractive. It can also provide financial returns in quick turnaround time (like harvest season). Ensuring mobility is a challenge for digital farming. Majorly digital farming in India has a better chance of being successful. This is because typical Indian farms are very small in size, and farm plots are of 1 to 2 acres. Moreover, hiring of a farming plot under various preparations is extensively experienced in India. From this, it is common to find out what crop a farmer is cultivating on a particular plot in one season and what the same farmer can transfer to another farm plot in the next season. In such a scenario, equipment that can be simply connected and reinstalled and used for multiple positions

during its lifespan would certainly be more attractive for the agriculturalist to pay and endow.

3.6 Future Scope and Benefits

There are many reasons to implement digital farming solutions in local and commercial farming. While the IoT in the world is fast-tracking the steps on data gathering and automation, significant industries like agriculture can certainly benefit through it. Soil wetness, air temperature, air moisture, and sunlight intensity can be monitored in many areas through digital technology. By collecting data through this, water use and crop yield efficiency in fields can also be improved in large and local farms. As the world's population grows over time, so will farming and food production. Therefore, the use of digital technology will enable low-cost sensors, data insights, and IoT platforms to increase efficiency and production. Advances in machinery have also expanded the scale, speed, and productivity of agricultural equipment. This has led to more productive cultivation in the land more efficiently. Seeds, irrigation, and fertilizers have also improved significantly, helping farmers to increase yields.

Following are some of the benefits of implementing digital farming solutions:

1. Water preservation: Weather forecasting and soil moisture sensors allow the use of water only when needed, which is a major part of farming.
2. To accelerate the decision-making process through digital farming, farmers can imagine fabrication levels, soil wetness, sunlight intensity, and more in real time as all these are key factors in farming.
3. Automated processes in planting, processing, and harvesting in digital farming reduce resource consumption, human error, and overall cost and lower operating costs, resulting in higher profits for farmers.
4. The quality of production can be increased with the use of digital farming. The results in relation to the analysis and treatment of production quality enable the farmers to adjust the processes to enhance the quality of the product. In addition, accurately estimating production rates by area over time can provide detailed estimates of future crop yields and farm value.
5. Through digital farming, all conservation efforts such as increasing water use and production per land unit directly impact the environmental footprint positively.
6. Many important uses of digital farming include monitoring and maintaining farm equipment to predict production rates, labor effectiveness, and failure predictions.

As we have seen in the above-mentioned benefits, farmers can actually obtain the solution or implementation through digital farming overall. It is not only a digital farming solution, but it is the innovative farming method of today. This innovative approach can address the growing concern of the global population's food consumption and environmental footprint. Agriculture in the future will all use sophisticated technologies such as robots, temperature and humidity sensors, aerial images, and GPS technology. This advanced equipment, robotic systems, and precision agriculture will help make farms more profitable, efficient, safe, and environmentally friendly. Apart from this, there should be some specific priorities for the future, namely:

1. Better data collection about digital technologies and digitization should be facilitated at the regional and population level.
2. Information about urban and rural areas should be differentiated by digital technologies so that it can be used according to urban and rural areas.
3. Involving small-scale farmers in the digital agriculture transformation process is feasible to build sustainable business models that provide digital solutions.
4. An index should be created that is based on the development of digital agriculture in the context of the cultural,

- educational, and institutional dimensions of a given country.
5. Digital agriculture should study the process in terms of both the basic conditions for digitization and the availability of enablers and the potential economic, social, and environmental impacts.

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Note

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